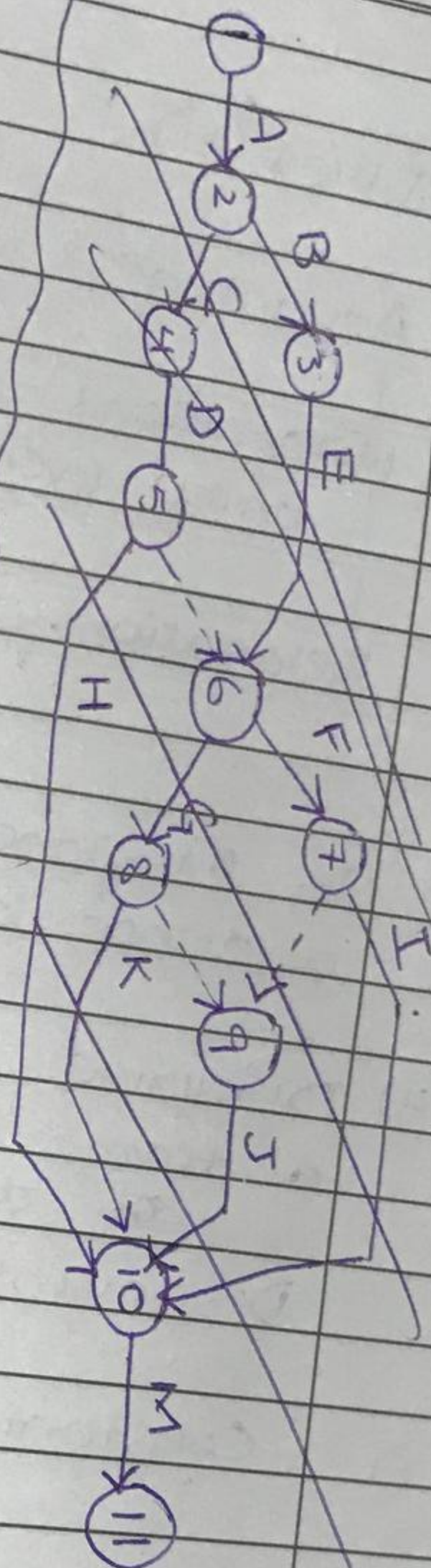
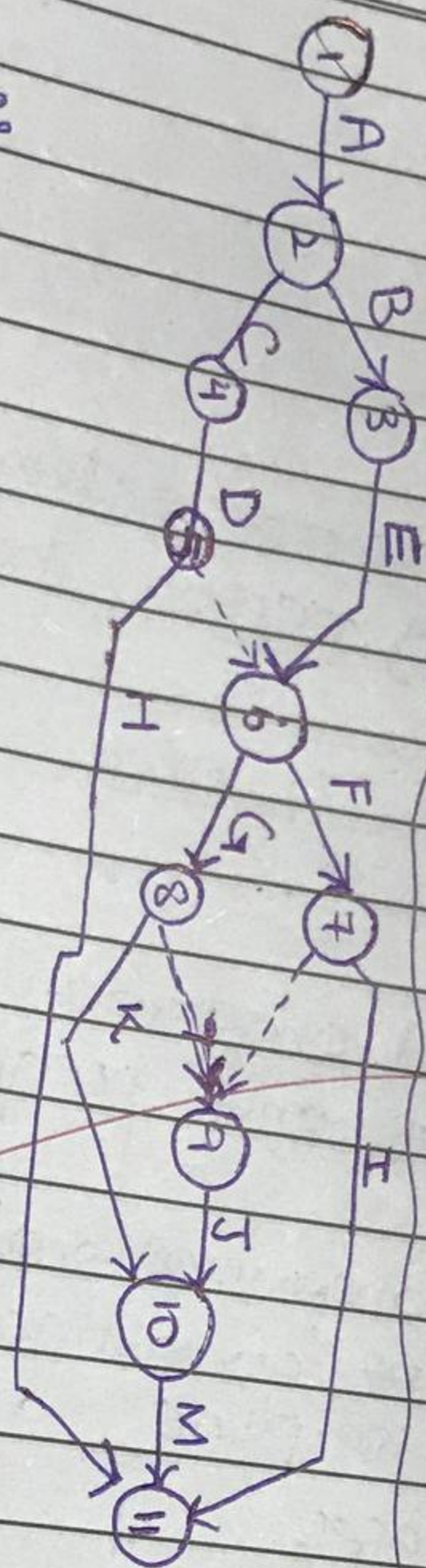


Neat Diagram :

8



Critical path :

A-B-C-D-G-K-M

Duration :

30 days

National University of Computer and Emerging Sciences, Lahore Campus



Course Name: Engineering Management
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 17-November-18
 Section: ALL
 Exam Type: Mid II

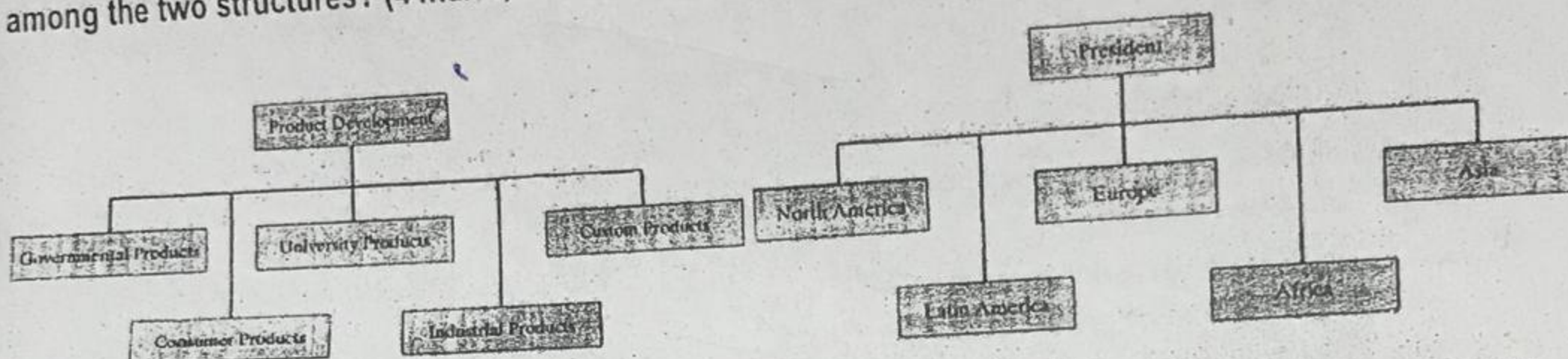
Course Code: MG 330
 Semester: Fall 2018
 Total Marks: 38
 Weight: 15%
 Page(s): 1

Student Name: Maida

Roll No. 15-4405 Section: EE-1

Instruction/Notes: Start each question from new page, show all steps, and assume if something is missing in paper. Return the question paper.

Q #1: (a) What kind of organizations are shown in the following two figure? What is ONE point of commonality among the two structures? (4 marks, CLO 5)



(b) List four advantages and four disadvantages of the above organizational structures. (4 marks, CLO 5)

Q #2: Answer the following multiple choice questions by encircling the correct answer on the question paper and return the question paper at the end of exam along with the answer sheet. (2 marks per MCQ; 20 Marks; CLO-4 & CLO-5)

A. Identify which one of the following is not action planning

- a. Develop a budget
- ☒ b. Build corporate know-how
- c. Define standards
- d. Analyze critical needs

B. What is not organizing?

- a. Defining team structure
- b. Defining team relationships
- c. Assigning product development activities to teams
- ☒ d. Identifying new markets

C. Which of the following is not a reason for failure of a strategic plan?

- a. Quality of strategic thinking
- b. Quality of communicating
- ☒ c. Having both internal and external focus
- d. Level of flexibility

D. Which of the following is the best definition for quality for a company?

- a. User based
- b. Product based
- c. Production based
- ☒ d. Value based

E. Which of the following is not a tool for planning?

- ☒ a. Technology IP
- b. Market research
- c. Performance bench mark
- d. Technology forecasting

F. Which organizational structure is most suited for a global company?

- a. Product based
- ☒ b. Region based
- c. Discipline based
- d. Matrix

G. Which of the following does not belong to Strategic Planning?

- a. Company's goals
- b. Company's revenue targets
- ☒ c. Company's product Design
- d. Company's profitability

H. Which of the following definition is wrong?

- a. Span of control is the number of people managed by a supervisor.
- b. A line-organization performs activities directly related to organization goals.
- c. Accountability is an upward-directed obligation to secure the desired results for the tasks assigned
- ☒ d. Responsibility is the duty to perform work due to accountability

I. Which of the following do not have a key role in strategic planning?

- ☒ a. Managers
- b. Senior Managers
- c. Directors
- d. Vice presidents

J. Which of the following does not relate to a type of a conflict

- a. *Technical*, including design, analysis, and interpretation of test results
- b. *operational*, including procedures to perform specific tasks and assign responsibility
- c. *Emotional*, such as treating bruised egos and hurt personal feelings
- ☒ d. None of the rest

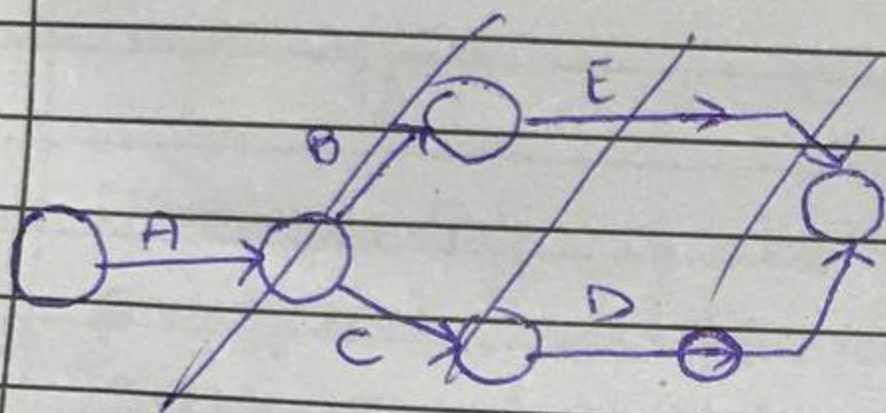
Q #3: Draw a neat network diagram. Find critical path and its duration. (marks 10, CLO-3)

Activity	A	B	C	D	E	F	H	I	G	J	K	L	M
Predecessor	-	A	A	C	B	D,E	D	F	D,E	F,G	G	H	K,J
Duration(days)	5	3	4	6	7	2	8	6	4	4	5	2	3

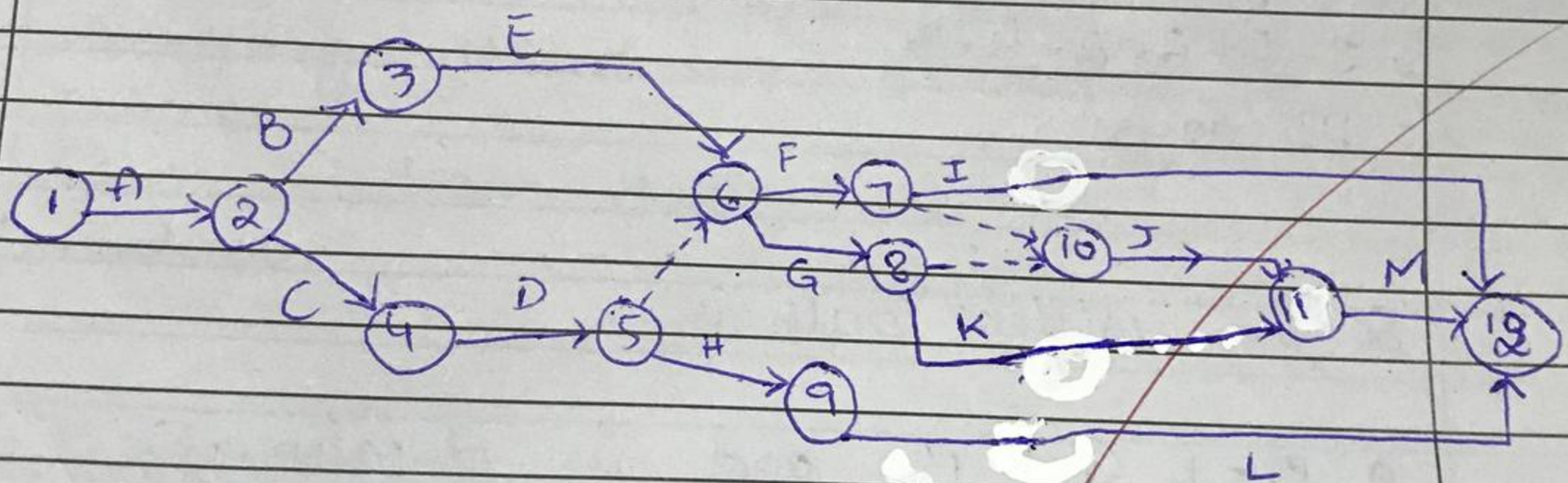
Course Engineering Management Answer Sheet No. 16146Student's Name Maida Signature C1 deRoll No. 15-4405 Section EE-1 Date 6-17-11-18

QUESTION 3

CRITICAL PATH AND NETWORK DIAGRAM



10



To compute critical path.

$$A \rightarrow B \rightarrow E \rightarrow F \rightarrow I \\ = 5 + 3 + 7 + 2 + 6 = 23 \text{ days}$$

$$A \rightarrow B \rightarrow E \rightarrow G \rightarrow J \rightarrow M \\ = 5 + 3 + 7 + 4 + 4 + 3 = 26 \text{ days}$$

$$A \rightarrow B \rightarrow E \rightarrow G \rightarrow K \rightarrow M \\ = 5 + 3 + 7 + 4 + 5 + 3 = 27 \text{ days}$$

Part No.

$$\begin{aligned} &A-B-E-F-J-M \\ &= 5+3+7+2+4+3 \\ &= 24 \text{ days} \end{aligned}$$

$$\begin{aligned} &A-C-D-H-L \\ &= 5+4+6+8+2 \\ &= 25 \text{ days} \end{aligned}$$

$$\begin{aligned} &A-C-D-F-I \\ &= 5+4+6+2+6 \\ &= 23 \text{ days} \end{aligned}$$

$$\begin{aligned} &A-C-D-G-J-M \\ &= 5+4+6+4+4+3 \\ &= 26 \text{ days} \end{aligned}$$

$$\begin{aligned} &A-C-D-F-J-M \\ &= 5+4+6+2+4+3 \\ &= 23 \text{ days} \end{aligned}$$

(e) So the critical path is

A-B-E-G-K-M and its duration is 27 days.